

The University of Zambia
Department of Mathematics and Statistics
Mat 1110 - Foundation Mathematics and Statistics for
Social Sciences

Tutorial Sheet 2- Sets of Numbers

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1. Express each of the following numbers in the form $\frac{a}{b}$, where $a, b \in \mathbb{Z}$, $b \neq 0$ and $\frac{a}{b}$ is in its simplest form.

(a) $0.222\ldots$

(d) $0.2121212121\ldots$

(g) -0.75

(b) $0.7\overline{3}$

(e) $-1.\overline{73}$

(c) $3.24\overline{15}$

(f) $0.0\overline{123}$

(h) $2.\overline{08}$

2. Let $A = \{-2, \frac{3}{4}, -\sqrt{5}, 0.7, \frac{2}{\sqrt{7}}, 0.\overline{37}, \sqrt{7}, 8.0, 9\}$. List the elements of the following subsets of A.

i. Natural numbers

ii. Integers

iii. Rational numbers

iv. Irrational numbers

v. Real numbers

3. (a) Let $X = (-10, 10]$ be the universal set and $A = (-2, 5)$, $B = [-5, 3]$ and $C = [-1, 8)$. Find each of the following sets and display it on the number line.

i. A'

ii. $X - A$

iii. $(A \cap C)'$

iv. $(B - A) \cap C$

v. $A \cap (B - C)$

vi. $(X - C)' \cap (A - B)$

- (b) Given that $\mathbb{R}$, is the set of real numbers is the universal set, $A = (-7, 2]$ and $B = [4, \infty)$, find

i. A'

ii. B'

iii. $A - B$

iv. $B - A$

4. Simplify each of the following:

(a) $7\sqrt{3} + \sqrt{12}$

(c) $\sqrt{20} + 2\sqrt{45} - \sqrt{80}$

(e) $\sqrt{108} - 8\sqrt{3} + 3\sqrt{48}$

(b) $\sqrt{200} + \sqrt{18} - \sqrt{72}$

(d) $3\sqrt{27} - \sqrt{48} + 2\sqrt{75}$

(f) $\frac{3\sqrt{28}}{2\sqrt{175}}$

5. Rationalize the denominator in each of the following:

(a) $\frac{3}{\sqrt{7}}$

(c) $\frac{3\sqrt{2}}{2 + \sqrt{5}}$

(e) $\frac{x - y}{\sqrt{x} - \sqrt{y}}$

(b) $\frac{2}{5 - \sqrt{2}}$

(d) $\frac{\sqrt{3} - \sqrt{5}}{\sqrt{3} + \sqrt{5}}$

(f) $\frac{h}{\sqrt{x+h} - \sqrt{x}}$

6. Rationalize the denominator, and simplify each of the following:

(a) $\frac{4}{\sqrt{2}} + \sqrt{8}$

(b) $\frac{8}{3 + \sqrt{5}} + \sqrt{45}$

(c) $\frac{\sqrt{3} - \sqrt{7}}{\sqrt{3} + \sqrt{7}} + \sqrt{21}$

7. Simplify each of the following leaving the answer in standard form, $a + bi$, where $a, b \in \mathbb{R}$.

(a) $(2 + 3i) - (-7 + 6i)$

(d) $(2 + i)(-1 + 2i)(4 - 3i)$

(f) $\frac{(-2 + 3i)^2}{1 + 2i}$

(b) $(3 + 4i) + (6 - 5i)$

(c) $(-2 + 3i)^2$

(e) $\frac{2 - 3i}{1 - 2i}$

8. Given that $z_1 = 3 - 2i$, $z_2 = -1 + i$ and $z_3 = 3 + 2i$, perform each of the following operations, leaving each answer in the form $a + bi$, where $a, b \in \mathbb{R}$.

(a) $\frac{2z_1 - 5z_3}{\bar{z}_3}$

(b) $\frac{\bar{z}_2 z_1}{z_3}$

(c) $\frac{(z_2)^2}{\bar{z}_1}$

9. Solve each of the following equations.

(a) $3 + 5i = (a + bi)(1 + i)$

(c) $(6 + 3i) = 4i - 30(x + yi)$

(b) $(a + i)^3 = 18 + 26i$

(d) $xyi = 3x - y + 12i$

10. Given that $\frac{5 - 2i}{z} = 2 + i$, find z in the form $a + bi$, where $a, b \in \mathbb{R}$.